

Assignment

Problem 1: Print numbers from 1 to 10 using a while loop.

Problem 2: Print even numbers between 1 and 50

Problem 3: Sum of first n natural numbers

Input: n = 10

Output: 55

Problem 4: Reverse a number

Input: 12345

Output: 54321

Problem 5: **Count digits in a number**

Input: 908765

Output: 6

Problem 6: Check if a number is a palindrome

Input: 121

Output: Palindrome

Problem 7: **Find factorial of a number**

Input: 5

Output: 120

Problem 8: **Print Fibonacci series up to n terms**

Input: n = 7

Output: 0 1 1 2 3 5 8

Problem 9: **Find sum of digits**

Input: 789

Output: 24

Problem 10: **Guess the number game**

Keep asking the user to guess a number until they guess correctly.

Problem 11: **ATM Withdrawal Simulation**

Keep withdrawing money until balance becomes 0 or less.

Input:

- Initial balance = 10,000
- Withdrawal per step = 1,500

Problem 12: **Data Quality Check**

Read values until an invalid value (-1) appears.
Count how many valid values were entered.

Problem 13: Streaming Sensor Data

Process sensor values until temperature crosses **80°C**.

Problem 14: Salary Increment Simulation

Start salary = 30,000

Increase by 10% every year

Stop when salary exceeds 1,00,000

Print number of years taken.

Problem 15: Find GCD using while loop

Input: 48, 18

Output: 6

Problem 16: Remove duplicate characters from string

Input: "analytics"

Output: "anlyts"

(No for loop allowed)

Problem 17: Read list until user enters "stop"

Store values and print final list.

Problem 18: Find prime numbers between 1 and 100 using while loop only

Problem 18: Remove vowels

Input: "analytics"

Output: "nlyts"

(No for loop allowed)

